

Analog-Verse (Ingenium) Literature review

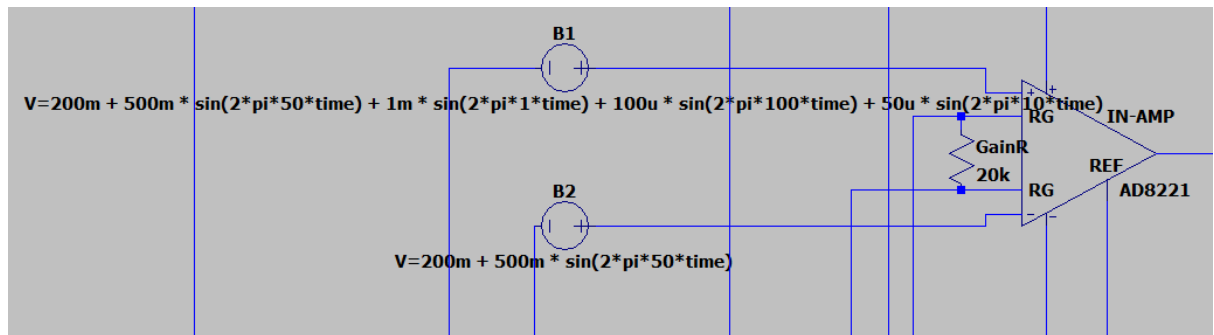
Team - ELECORE

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The Task: To isolate and return alpha brain waves using a 3-electrode analog-only, front end wearable device.

Topologies Used:

1. An Instrumentation Amplifier (In-Amp)-



The instrumentation amplifier is used over a regular op-amp as it provides a very high CMRR usually over 100dB and is better for rejecting noise. The AD8221 specifically has a CMRR of between 100-130dB depending on the frequency of input signal.

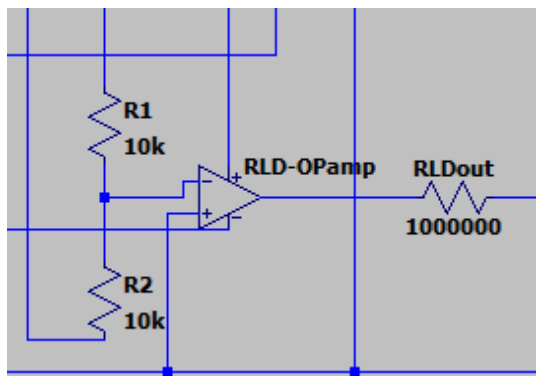
It also has a single resistor R_g for adjusting the gain which is given by

$$G = 1 + (49.4\text{k}\Omega / R_g) \quad \text{for the AD8221.}$$

For the shown arrangement the gain is 33.96.

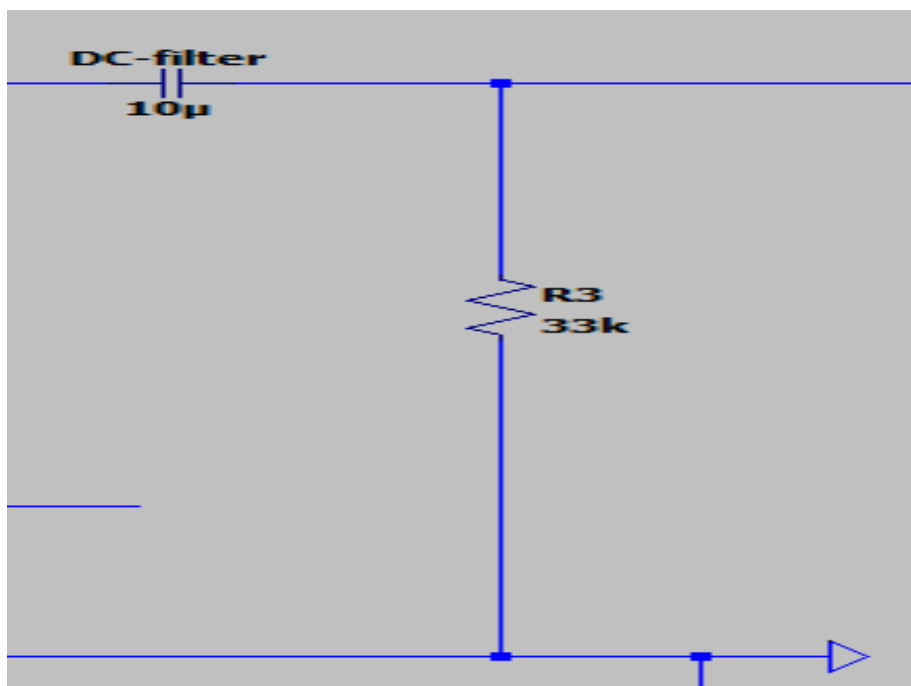
The in-amps also have very high input impedance which means that none of the wave voltage gets dropped across the electrode (electrode loading).

2. The Right Leg Drive-



The Right Leg Drive is the third electrode generally connected to the human's right leg. It takes the potential across the input (common mode voltage) through 2 10kΩ resistors, inverts it through the op-amp and sends it back to the body through a high resistance (for safety). It therefore cancels the 50Hz noise up to an extent in the body.

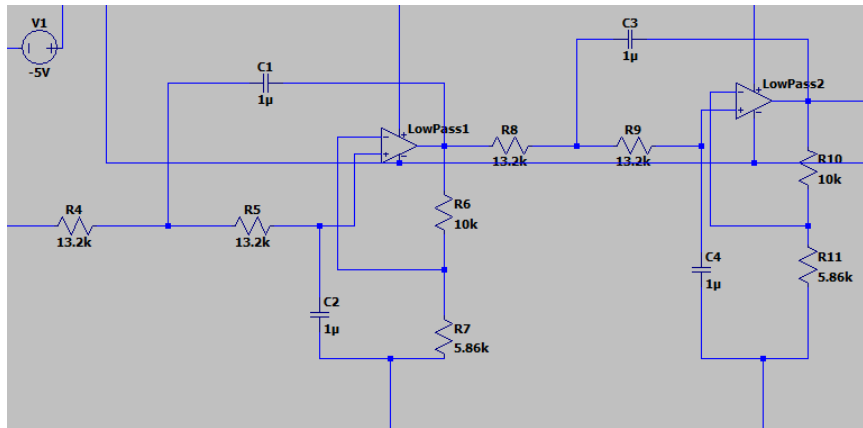
3. The DC Filter:



This is a simple high pass filter with a cutoff frequency of 0.48Hz. It ensures that no DC current flows or gets amplified through the following active filters preventing power rail saturation.

4. Low and High Pass Sallen Key filters-

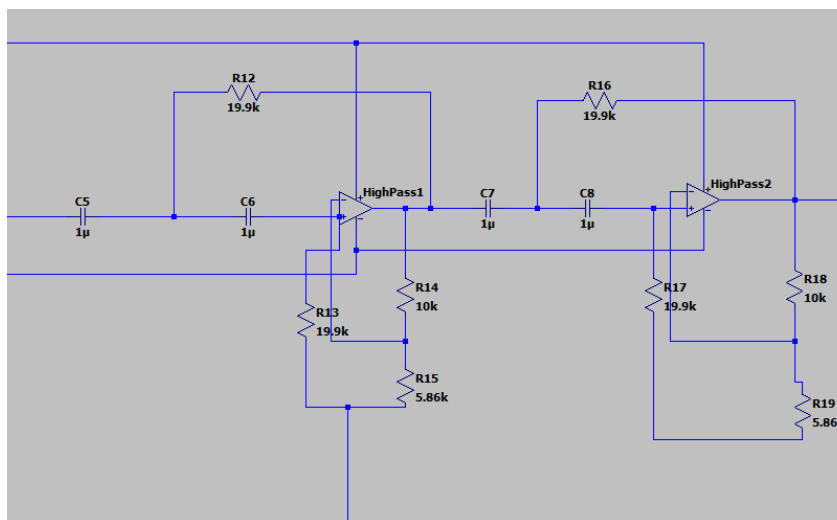
The Sallen-Key topology is used over MFB as it has high input impedance and is fed directly into the non-inverting input of the op-amp. The gain in the Sallen-key topology is also independent of the frequency characteristics of the filter.



This is the low pass Sallen-Key filter. This is a second order active filter. The order is increased to 4 by cascading 2 low pass filters. The order is increased in order to provide a sharp roll-off of 80dB per decade to create a plateau between 8-12Hz in the Bode plot.

The system gain can be adjusted by changing R6, R7, R10 or R11 and the gain for each stage is given by $(1+R_f/R_g)$ where R_f is R6 or R10 and R_g is R7 or R11.

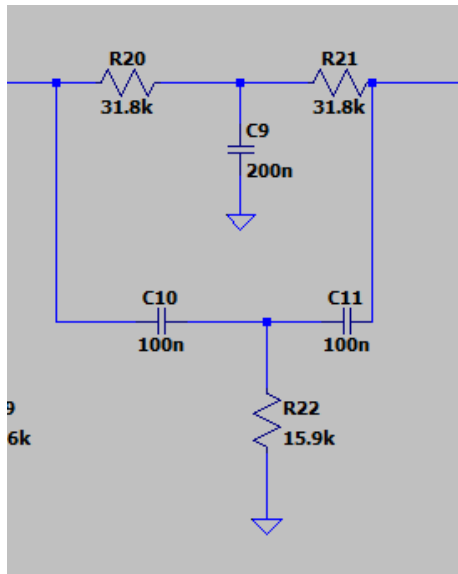
The cutoff frequency for this filter is 12Hz.



This is the high pass Sallen-Key filter. It has the same gain as the previous stages and has a cutoff frequency of 8Hz.

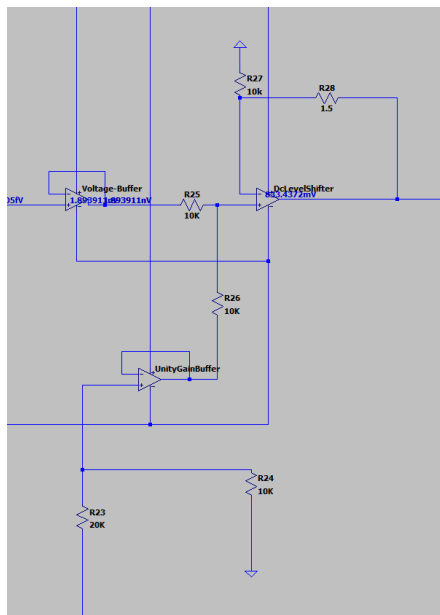
On cascading the above 2 filters we get a pass band between 8 and 12 Hz with a surrounding roll-off of 80dB per decade.

5. The Twin-T Notch Filter:



This filter is used in order to isolate the 50Hz frequency mains hum and filter it out. In the Bode plot a notch is seen at the 50Hz point ideally forming a pass band above and below 50Hz.

6. The DC Level Shifter

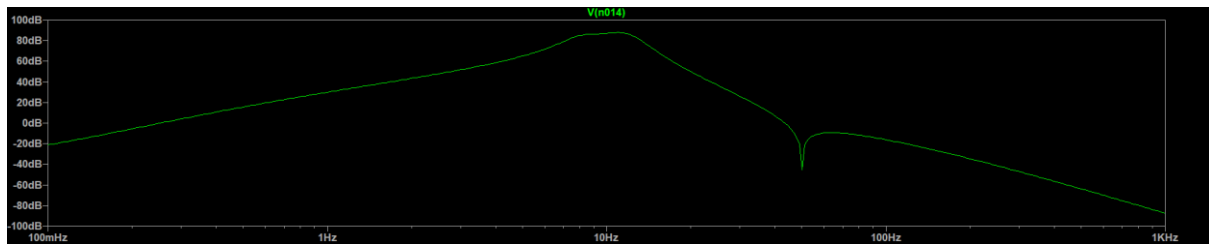


Unity-gain buffers act as one-way walls between different parts of the circuit, preventing the stages from interfering with each other. The DC level shifter then takes the AC brainwave and

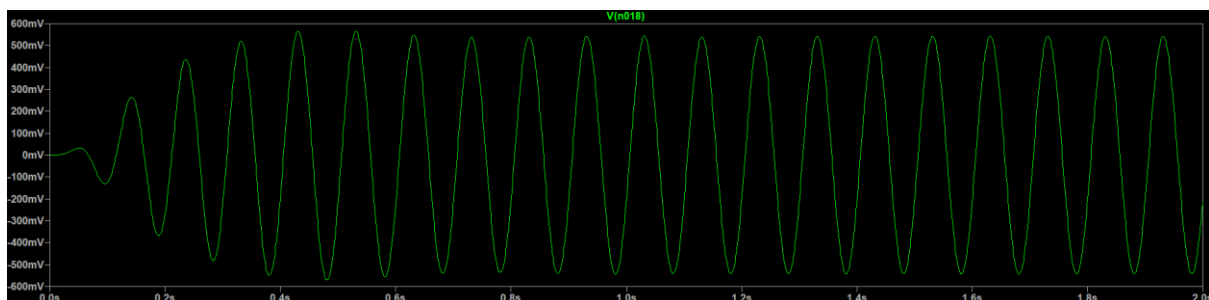
pushes it up to a positive resting voltage. This is necessary because microcontrollers cannot read negative voltages, so shifting the signal up prevents the bottom half of the wave from getting cut off.

Output Characteristics and Logs:

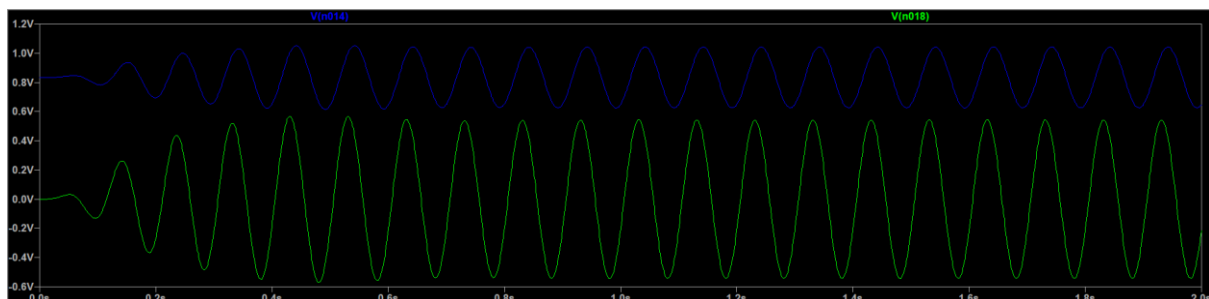
1. The output Bode plot(Test 11) with a plateau between 8-12Hz and a visible notch at 50Hz. (AC analysis)



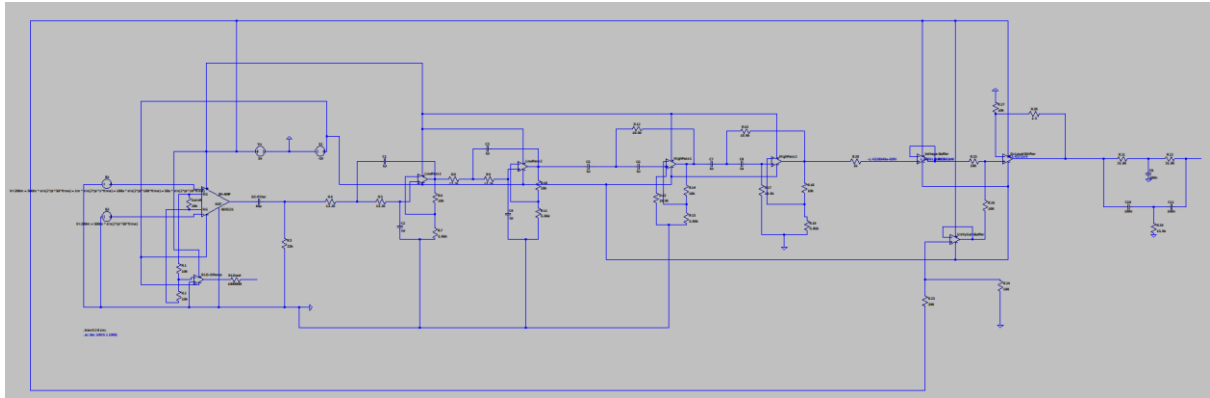
2. The output plot for transient analysis (Test 10) showing a clean 10Hz signal before the level shifting.



3. The output plot for the final wave ready to be fed into an ADC after level shifting.



Complete Circuit:



We've added $\pm 5V$ DC source and the total system gain is 5000.